Category
Introduction to Big Data Techniques

Related Questions:
Practice questions related to this topic

TACKLING BIG DATA WITH DATA SCIENCE

Learning Outcome

- describe applications of Big Data and Data Science to investment management

Data science can be defined as an interdisciplinary field that harnesses advances in computer science (including ML), statistics, and other disciplines for the purpose of extracting information from Big Data (or data in general). Companies rely on the expertise of data scientists/analysts to extract information and insights from Big Data for a wide variety of business and investment purposes.

An important consideration for the data scientist is the structure of the data. As noted in the discussion on Big Data, because of their unstructured nature, alternative data often require specialized treatment before they can be used for analysis.

Data Processing Methods

To help determine the best data management technique needed for Big Data analysis, data scientists use various data processing methods, including capture, curation, storage, search, and transfer.

- Capture—Data capture refers to how the data are collected and transformed into a format that can be used by the analytical process. Low-latency systems—systems that operate on networks that communicate high volumes of data with minimal delay (latency)—are essential for automated trading applications that make decisions based on real-time prices and market events. In contrast, high-latency systems do not require access to real-time data and calculations.
- Curation—Data curation refers to the process of ensuring data quality and accuracy through a data cleaning exercise. This process consists of reviewing all data to detect and uncover data errors—bad or inaccurate data—and making adjustments for missing data when appropriate.
- Storage—Data storage refers to how the data will be recorded, archived, and accessed and the underlying database design. An important consideration for data storage is whether the data are structured or unstructured and whether analytical needs require low-latency solutions.
- Search—Search refers to how to query data. Big Data has created the need for advanced applications capable of examining and reviewing large quantities of data to locate requested data content.
- Transfer—Transfer refers to how the data will move from the underlying data source or storage location to the underlying analytical tool. This could be through a direct data feed, such as a stock exchange's price feed.

Data Visualization

Data visualization is an important tool for understanding Big Data. Visualization refers to how the data will be formatted, displayed, and summarized in graphical form. Traditional structured data can be visualized using tables, charts, and trends, whereas non-traditional unstructured data require new techniques of data visualization. These visualization tools include, for example, interactive three-dimensional (3D) graphics, in which users can focus in on specified data ranges and rotate the data across 3D axes to help identify trends and uncover relationships. Multidimensional data analysis consisting of more than three variables requires additional data visualization techniques—for example, adding color, shapes, and sizes to the 3D charts. Furthermore, a wide variety of solutions exists to reflect the structure of the data through the geometry of the visualization, with interactive graphics allowing for especially rich possibilities. Examples include heat maps, tree diagrams, and network graphs.

Another valuable Big Data visualization technique that is applicable to textual data is a "tag cloud," in which words are sized and displayed on the basis of the frequency of the word in the data file. For example, words that appear more often are shown with a larger font, and words that appear less often are shown with a smaller font. A "mind map" is another data visualization technique; it is a variation of the tag cloud, but rather than displaying the frequency of words, a mind map shows how different concepts are related to each other.

Exhibit 3 shows an example of a "tag cloud" based on a section of this reading. The more frequently a word is found within the text, the larger it becomes in the tag cloud. As shown in the tag cloud, the words appearing most frequently in the section include "data," "ML," "learning," "AI," "techniques," "model," and "relationships."

Exhibit 3: Data Visualization Tag Cloud

Source: "About Word Clouds," WordItOut, https://worditout.com/word-cloud/create.

Fintech is being used in numerous areas of investment management. Applications for investment management include text analytics and natural language processing, risk analysis, and algorithmic trading.

Text Analytics and Natural Language Processing

Text analytics involves the use of computer programs to analyze and derive meaning typically from large, unstructured text- or voice-based datasets, such as company filings, written reports, quarterly earnings calls, social media, email, internet postings, and surveys. Text analytics includes using computer programs to perform automated information retrieval from different, unrelated sources to aid the decision-making process. More analytical usage includes lexical analysis, or the analysis of word frequency in a document and pattern recognition based on key words and phrases. Text analytics could be used in predictive analysis to help identify indicators of future performance, such as consumer sentiment.

Natural language processing (NLP) is a field of research at the intersection of computer science, AI, and linguistics that focuses on developing computer programs to analyze and interpret human language. Within the larger field of text analytics, NLP is an important application. Automated tasks using NLP include translation, speech recognition, text mining, sentiment analysis, and topic analysis. NLP also might be used in compliance functions to review employee voice and electronic communications for adherence to company or regulatory policy, for detecting fraud or inappropriate conduct, or for ensuring private or customer information is kept confidential.

Consider that all the public corporations worldwide generate millions of pages of annual reports and tens of thousands of hours of earnings calls each year. This is more information than any individual analyst or team of researchers can assess. NLP, especially when aided by ML algorithms, can analyze annual reports, call transcripts, news articles, social media posts, and other text- and audio-based data to identify trends in shorter time spans and with greater scale and accuracy than is humanly possible.

For example, NLP can be used to monitor analyst commentary to aid investment decision making. Financial analysts might generate earnings-per-share (EPS) forecasts reflecting their views on a company's near-term prospects. Focusing on forecasted EPS numbers could mean investors miss subtleties contained in an analyst's written research report. Because analysts tend not to change their buy, hold, and sell recommendations for a company frequently, they might instead offer nuanced commentary without making a change in their investment recommendation. After analyzing analyst commentary, NLP can assign sentiment ratings ranging from very negative to very positive for each. NLP, therefore, can be used to detect, monitor, and tag shifts in sentiment, potentially ahead of an analyst's recommendation change. Machine capabilities enable this analysis to scale across thousands of companies worldwide, performing work previously done by humans.

Similarly, communications and transcripts from policy makers, such as the European Central Bank or the US Federal Reserve, offer an opportunity for NLP-based analysis because officials at these institutions might send subtle messages through their choice of topics, words, and inferred tone. NLP can analyze nuances within text to provide insights around trending or waning topics of interest, such as interest rate policy, aggregate output, or inflation expectations.

Models using NLP analysis might incorporate non-traditional information to evaluate what people are saying —through their preferences, opinions, likes, or dislikes—in an attempt to identify trends and short-term indicators—for example about a company, a stock, or an economic event—to forecast coming trends that may affect investment performance in the future. For example, past research has evaluated the predictive power of Twitter sentiment regarding initial public offering (IPO) performance as well as the effect of positive and negative news sentiment on stock returns.

Programming Languages and Databases

Some of the more common programming languages used in data science include the following:

- **Python:** Python is an open-source, free programming language that does not require an in-depth understanding of computer programming. Python allows individuals with little or no programming experience to develop computer applications for advanced analytical use and is the basis for many fintech applications.

- **Java:** Java is a programming language that can run on different computers, servers, and operating systems. Java is the underlying program language used in many internet applications.
- **C and C++:** Both C and C++ are specialized programming languages that provide the ability to optimize source code to achieve superior calculation speed and processing performance. C and C++ is used in applications for algorithmic and high-frequency trading.
- **Excel VBA:** Excel VBA helps bridge the gap between programming and manual data processing by allowing users to run macros to automate tasks, such as updating data tables and formulas, running data queries and collecting data from different web locations, and performing calculations. Excel VBA allows users to develop customized reports and analyses that rely on data that are updated from different applications and databases.

Some of the more common types of databases in use include the following:

- **SQL:** SQL is a database query language for structured data where the data can be stored in tables with rows and columns. SQL-based databases need to be run on a server that is accessed by users using SQL queries.
- **SQLite:** SQLite is a database for structured data. SQLite databases are embedded into the program and do not need to be run on a server. It is the most common database for mobile apps that require access to data.
- **NoSQL:** NoSQL is a database used for unstructured data where the data cannot be summarized in traditional tables with rows and columns.

Rate Your Confidence

Category
Introduction to Big Data Techniques

Related Questions:
Practice questions related to this topic

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{NI AI} NEWS FEED ARTIFICIAL INTELLIGENCE

The Bloomberg NI AI function allows you to read the latest news on artificial intelligence in finance and business.

> "NI <GO> is a menu of news categories, which allows you to quickly search and browse for news stories corresponding to companies, regions, industries, and other topics maintained by Bloomberg, such as business or general news." -Bloomberg

To explore this function, type NI AI into the Bloomberg command line and press Enter or <GO>.

© 2023 Bloomberg L.P.

Please provide feedback on this Bloomberg feature.

Discuss
Filter
Discussion
What do you want to discuss... (required)

SK In the practice questions there were also questions related to "algorithmic trading" and "Risk governance", these however are not mentioned in this lesson. Is it part of the curriculum for 2025 but not mentioned in the lesson? Or it is left out of the curriculum for 2025, but not updated in the practice questions?

Created 3 months ago by Stefan Kuruc

1 reply | Last Activity: 3 months ago

Show All replies
Reply to this Comment

Rate Your Confidence

Category

Introduction to Big Data Techniques

Related Questions:

Practice questions related to this topic